

Lecture 17

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1 Multivariate Polynomial Codes

We define a multivariate polynomial code $\mathcal{C}$ over a binary field (i.e. $GF(2)$) as

$$\mathcal{C}(r, m) := \{(f(\alpha_1), f(\alpha_2), f(\alpha_3), \dots, f(\alpha_{2^m})) \mid f \in \mathbb{F}_2[X_1, X_2, X_3, \dots, X_m], \deg(f) \leq r\}$$

Similar to univariate polynomials, we represent $f(X)$ as,

$$f(X) = \sum_{\substack{S \subseteq [m] \\ |S| \leq r}} c_S \prod_{i \in S} X_i \quad \text{where, } c_S \in GF(2)$$

The coefficients c_S correspond to the message $\mathbf{u}$.

Example: Consider the codeword belonging to $\mathcal{C}(2, 2)$ generated by the multivariate polynomial: $f(X) = 1 + X_1X_2$

At the evaluation points $\{00, 01, 10, 11\}$, we get:

$$f(00) = 1$$

$$f(01) = 1$$

$$f(10) = 1$$

$$f(11) = 0$$

Thus the codeword generated is 1110.

Since $X_i^n = X_i$, we only consider the presence (X_i^1) or absence (X_i^0) of X_i in any term.

1.1 Code Parameters

Since all X_i 's in X can either take the value 0 or 1, we have a total of 2^m assignments. Thus, the length of the codeword evaluated for all these assignments will be 2^m .

$$n = 2^m$$

If we have a message $(u_0, u_1, \dots, u_{k-1})$, then we can map each message bit to a particular c_S . Thus the number of message bits (k) is equal to the number of total possible subsets S .

$$k = 1 + \binom{m}{1} + \binom{m}{2} + \binom{m}{3} + \dots + \binom{m}{r}$$

Lemma 1 (Schwartz-Zippel Lemma) *Let $f \in \mathbb{F}_2[X_1, X_2, \dots, X_m]$ be a non-zero polynomial such that $\deg(f) \leq r$, then:*

$$\sum_{\alpha \in \mathbb{F}_2^m} \mathbb{1}(f(\alpha) \neq 0) \geq 2^{m-r}$$

which is also represented as

$$\Pr_{\alpha \in \mathbb{F}_2^m} [f(\alpha) \neq 0] \geq 2^{-r}$$

The bound that we get from Schwartz-Zippel Lemma is also tight i.e. we can find a f which has exactly 2^{m-r} non-roots (ex. $f(X) = \prod_{i=1}^r X_i$. This will have non-roots when X_1 to X_r are all 1's, and since the remaining $(m-r)$ X_i 's can be either 0 or 1, we get exactly 2^{m-r} non-roots).

Thus, we can see that the minimum distance of our code $\mathcal{C}$ is 2^{m-r}

$$d = 2^{m-r}$$

It turns out that the (n, k, d) parameters for the Multivariate Polynomial codes are exactly the same as those for the Reed-Muller codes over a given finite field. The code $\mathcal{C}(r, m)$ can thus be seen as an alternate multi-variate description of a Reed-Muller code.

General description of the Schwartz-Zippel Lemma over $\mathbb{F}_q$

Lemma 2 *Let $f \in \mathbb{F}_q[X_1, X_2, \dots, X_m]$ be a non-zero polynomial with total degree at most d , then:*

$$\Pr_{\alpha \in \mathbb{F}_q^m} [f(\alpha) \neq 0] \geq \frac{1}{q^{\lfloor \frac{d}{q-1} \rfloor}} \left(1 - \frac{d \bmod (q-1)}{q} \right)$$

2 Decoding Rule

For any $S \subseteq [m]$ we define $R_S(X) := \prod_{i \in S} X_i$. Thus, we can write

$$f(X) = \sum_{\substack{S \subseteq [m] \\ |S| \leq r}} c_S R_S(X)$$

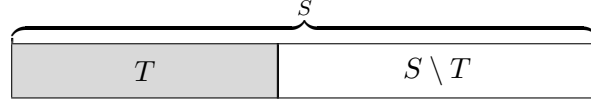
Lemma 3 *For any set S*

$$\sum_{\alpha \in \mathbb{F}_2^{|S|}} R_S(\alpha) = 1$$

Proof Only for one value of α , where every element is 1 (11...1) will $R_S(\alpha)$ evaluate to 1, all of the remaining values give 0. Hence summation of $R_S(X)$ over all α gives 1. $\blacksquare$

Lemma 4 *For any set S , for some $T \subset S$*

$$\sum_{\alpha \in \mathbb{F}_2^{|S|}} R_T(\alpha) = 0$$



Proof For some $i \in S \setminus T$,

$$\sum_{\alpha \in \mathbb{F}_2^{|S|}} R_T(\alpha) = \sum_{\alpha: \alpha_i=0} R_T(\alpha) + \sum_{\alpha: \alpha_i=1} R_T(\alpha)$$

Since $i \notin T$, R_T is independent of X_i . Hence the value of the summation remains the same regardless of α_i .

$$\sum_{\alpha: \alpha_i=0} R_T(\alpha) = \sum_{\alpha: \alpha_i=1} R_T(\alpha) = x \text{ (Let)}$$

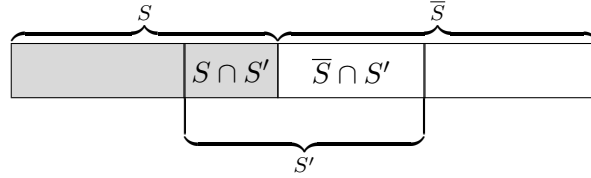
Thus,

$$\sum_{\alpha \in \mathbb{F}_2^{|S|}} R_T(\alpha) = x + x = 0$$

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Lemma 5 For any set S such that $|S| = r$ and $b \in \mathbb{F}_2^{m-r}$

$$\sum_{\substack{X_S = \alpha \in \mathbb{F}_2^m \\ X_{\bar{S}} = b}} f(X) = c_S$$



Proof Define $f_b(\alpha)$, as $f(X)$ where $X_S = \alpha$ and $X_{\bar{S}} = b$, i.e. the values of $X_i \forall i \in \bar{S}$ are fixed based on b , and the remaining X_i are varied with α .

For any $S' \subseteq [m]$, either $S' = S$ or $S' \neq S$.

For $S' = S$, its term will remain as-is in the expansion of $f_b(\alpha)$ i.e. $c_S R_S(X)$. However, when $S' \neq S$,

$$\begin{aligned} c_{S'} R_{S'}(X) &= c_{S'} \prod_{i \in S'} X_i \\ &= c_{S'} \prod_{i \in S' \cap \bar{S}} X_i \cdot \prod_{i \in S' \cap S} X_i \\ &= \left(c_{S'} \prod_{i \in S' \cap \bar{S}} X_i \right) \cdot R_{S' \cap S}(X) \\ &= k R_{S' \cap S}(X) \end{aligned}$$

As $S' \neq S$, $S' \cap S \subset S$. Therefore,

$$\begin{aligned}
f_b(\alpha) &= c_S R_S(\alpha) + \sum_{T \subset S} a_T R_T(\alpha) \\
\sum_{\alpha \in \mathbb{F}_2^m} f_b(\alpha) &= \sum_{\alpha \in \mathbb{F}_2^m} \left(c_S R_S(\alpha) + \sum_{T \subset S} a_T R_T(\alpha) \right) \\
&= c_S \sum_{\alpha \in \mathbb{F}_2^m} R_S(\alpha) + \sum_{T \subset S} \left(a_T \sum_{\alpha \in \mathbb{F}_2^m} R_T(\alpha) \right) \\
&= c_S \cdot 1 + \sum_{T \subset S} 0 \quad \because \text{Lemma 3 \& 4} \\
&= c_S
\end{aligned}$$

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3 Decoding Rule

As per Lemma 5, for any S we can fix b in 2^{m-r} possible ways i.e. we have 2^{m-r} equations for finding the value of c_S .

For a received output $\mathbf{r}$, evaluate the 2^{m-r} parity checks corresponding to S . For error correction, employ majority decoding as an estimate for c_S . As such, we can correct for $\lfloor \frac{d-1}{2} \rfloor = 2^{m-r-1} - 1$ errors.

Example: Consider a code-word from $\mathcal{C}(1, 3)$ generated by $f(X) = 1 + X_1 + X_2$, transmitted over a channel and received as $\mathbf{r}$.

With the evaluation points $\{000, 001, 010, 100, 110, 101, 011, 111\}$:

X	f(X)	$\mathbf{r}$
000	$f(000) = 1$	0
001	$f(001) = 1$	1
010	$f(010) = 0$	0
100	$f(100) = 0$	0
110	$f(110) = 1$	1
101	$f(101) = 0$	0
011	$f(011) = 0$	0
111	$f(111) = 1$	1

Let $S = \{1\}$, then $\overline{S} = \{2, 3\}$

$$\text{For } b = 00, r(000) + r(100) = 0 + 0 = 0$$

$$\text{For } b = 01, r(001) + r(101) = 1 + 0 = 1$$

$$\text{For } b = 10, r(010) + r(110) = 0 + 1 = 1$$

$$\text{For } b = 11, r(011) + r(111) = 0 + 1 = 1$$

As per majority decoding, we assign $c_{\{1\}} = 1$. Similarly, we get $c_{\emptyset} = 1$, $c_{\{2\}} = 1$ and $c_{\{3\}} = 0$. Thus, we recover the original message of 1110.